4.4 Parts of a Circle

Circles are geometric shapes you can see all around you. The significance of the concept of a circle can be well understood from the fact that the wheel is one of the ground-breaking inventions in the history of mankind.



Fig. 4.43

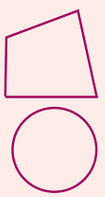
A circle, you can describe, is the set of all points in a plane at a constant distance from a fixed point. The fixed point is the centre of the circle; the constant distance corresponds to a radius of the circle.

A line that cuts the circle in two points is called a secant of the circle.

A line segment whose end points lie on the circle is called a chord of the circle.

Note

A circle notably differs from a polygon. A polygon (for example, a quadrilateral) has edges and corners while, a circle is a 'smooth' curve.



A chord of a circle that has the centre is called a diameter of the circle. The circumference of a circle is its boundary. (We use the term perimeter in the case of polygons).

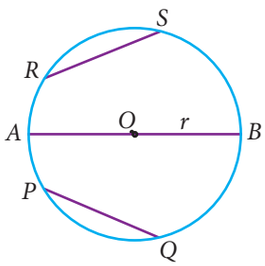


Fig. 4.44

In Fig.4.44, we see that all the line segments meet at two points on the circle. These line segments are called the chords of the circle. So, a line segment joining any two points on the circle is called a chord of the circle. In this figure AB, PQ and RS are the chords of the circle.

Now place four points P, R, Q and S on the same circle (Fig.4.45), then PRQ and QSP are the continuous parts (sections) of the circle. These parts (sections) are to be denoted by $\overset{\frown}{PRQ}$ and $\overset{\frown}{QSP}$ or simply by $\overset{\frown}{PQ}$ and $\overset{\frown}{QP}$. This continuous part of a circle is called an arc of the circle. Usually the arcs are denoted in anti-clockwise direction.

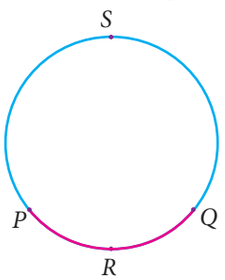


Fig. 4.45

Now consider the points P and Q in the circle (Fig.4.45). It divides the whole circle into two parts. One is longer and another is shorter. The longer one is called major arc $\overset{\frown}{QP}$ and shorter one is called minor arc $\overset{\frown}{PQ}$.

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